



**LUND**  
UNIVERSITY

**Written Examination**  
**Linear Analysis, MATB24**  
**August 20, 2021**  
**Time: 09.00–16.00**

Centre for Mathematical Sciences  
Mathematics, Faculty of Science

*Hand in electronically in the form of a PDF-file no later than at 15.59. You may freely use formulas from the formula sheet on the canvas page. Refer to relevant theorems from the course when you use them. Write clearly with short and concise motivations.*

1. Find a power series solution to the problem

$$y'' + xy' + y = 0, \quad y(0) = 1, y'(0) = 0.$$

Also determine the radius of convergence.

2. Consider the series  $s(x) = \sum_{n=1}^{\infty} \frac{x}{1+n^2x}$ .
- (a) Prove that  $s(x)$  is continuous for  $x > 0$ .
  - (b) Prove that  $s(x)$  continuously differentiable for  $x > 0$ .
  - (c) Prove that  $s(x)$  is strictly increasing for  $x > 0$ . What can be said about the limit  $\lim_{x \rightarrow 0} s(x)$ ?
3. Find a solution  $u(x, t)$  to the wave equation  $u_{tt} = 4u_{xx}$  when  $t > 0$ ,  $0 < x < \pi$  such that the boundary conditions  $u(0, t) = u(\pi, t) = 0$  ( $t > 0$ ) and the initial conditions  $u(x, 0) = \sin x$  and  $u_t(x, 0) = x(\pi - x)$  ( $0 < x < \pi$ ) are satisfied.
4. Compute the Fourier transform of  $f(x) = e^{-|x|} \sin |x|$  and use the result to compute the integral

$$\int_0^{\infty} \frac{(x^2 - 2)^2}{(x^4 + 4)^2} dx.$$

5. Let  $f(x)$  be the even,  $2\pi$ -periodic function which satisfies  $f(x) = \cos(2x)$  when  $0 \leq x \leq \frac{\pi}{2}$  and  $f(x) = -1$  when  $\frac{\pi}{2} \leq x \leq \pi$ . Calculate the Fourier coefficients and use the result to determine the value of the sum

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)(2n-1)(2n+3)}.$$

6. (a) Define what it means that a series  $\sum_1^{\infty} a_k$  be Cesàro summable to a value  $\sigma$ . Below, the value of the Cesàro sum is denoted  $\sigma = \sum a_k(C, 1)$ .
- (b) Prove that the series  $\sum_1^{\infty} \frac{(-1)^{n-1}}{n}$  is convergent (in the ordinary sense). Also try to find the sum of the series. (Hint: one can use the identity  $(1+t)^{-1} = \sum_0^{\infty} (-1)^n t^n$  for  $|t| < 1$ .)
- (c) Compute the Cesàro sum

$$\sigma = \sum_1^{\infty} (-1)^{n-1} \left(1 + \frac{1}{n} + \frac{(-1)^n}{n(n+1)}\right) (C, 1)$$

or show that it diverges.